DS-A1 — Data Provenance and Measurement Audit

NASA/GSFC SPDF OMNI 2 Daily Averages

Testable question. Within the frozen daily OMNI records, is a more southward daily mean IMF Bz (GSM Bz ≤ –5 nT) associated with higher daily Kp than days with Bz > –5 nT?

Provenance and legal usability. The dataset is the public OMNI 2 daily-average product. NASA SPDF states that, unless otherwise noted, its archived data and related materials are public domain and released under CC0. The OMNI documentation identifies the daily file as omni_01_av.dat and explains that daily values are arithmetic averages of hourly OMNI2 values.

Frozen version. Local file size 7,667,328 bytes; SHA-256 `1dd1327e601713be59903b064dbed45421673040f443f4148fd11a4deb4e0c1c`. The analysis uses this local copy, not a live download.

Audit item	Result
Rows × fields	23,376 × 55
Duplicate rows	0
Duplicate dates	0
Invalid calendar dates	0
NaN-only missing cells	0 (failed naive check)
Bz GSM sentinel missing	3,076 (13.16%)
Kp sentinel missing	106 (0.45%)

Data-generating process and lineage

Spacecraft/instruments → source measurements → OMNI cross-normalization and time shifting where applicable → hourly OMNI2 → arithmetic daily aggregation → frozen local file → parsing → sentinel-aware audit → association analysis.

The unit is one day. Population is the set of daily records represented by the frozen file; the analysis sample is the subset with non-fill Bz GSM and Kp. Stakeholder: a space-weather data user evaluating fitness for this association question. Target: daily Kp. Exposure: daily mean Bz GSM. Estimand: difference in mean standard Kp between Bz ≤ –5 nT and Bz > –5 nT.

Execution results

Group	n	Mean Kp	Median	SD
Bz <= -5 nT	266	4.636	4.7	1.269
Bz > -5 nT	20,034	2.041	2.0	1.123

Mean difference (Bz ≤ –5 minus other) = **2.595** standard Kp units. Pearson correlation between daily Bz GSM and standard Kp = **-0.328**.

Counterexample and limits

A short intense southward-Bz interval can be diluted by daily averaging, causing a day with an important sub-daily event to have a daily mean above –5 nT. Kp is itself a derived and rounded index. Therefore the threshold is a measurement choice, not a perfect event detector. The analysis does not establish causality, local-station effects,

sub-daily storm intensity, or forecasting performance. Using same-day Kp as a predictor of same-day Kp would also be leakage; this project uses same-day variables only for a descriptive association estimand.

AI-use and verification

ChatGPT assisted with question framing, notebook/report scaffolding, and audit checklist design. Independent verification consisted of hashing the local file, matching the 55-field schema to NASA documentation, checking sentinel values, duplicates, calendar validity, ranges, and recomputing all reported statistics from the frozen file in executable code. No observations were fabricated.

Source citation: Papitashvili, N. E. & King, J. H. (2020), OMNI Daily Data, NASA Space Physics Data Facility, DOI 10.48322/5fmx-hv56.